

Greater value add from electronic health records than polygenic risk scores for predicting myocardial infarction in machine learning

Corresponding Author: Mr Andrew Hornback

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

In this article, the authors investigate and compare the predictive value of polygenic risk scores, established cardiovascular risk scores and extended EHR-derived features in logistic regressions and feed-forward neural networks for 10-year risk of heart attacks in the UK-Biobank. More specifically, they find that while PRS are easily outperformed on their own by clinical and EHR-features across model types, they do add low but significant value in combination with them, and more so for the smaller clinical feature set than the larger EHR-based one. They further investigate which of these features are most important, and how they interact with each other.

While the authors address the highly relevant question of PRS utility in the light of different machine learning models and availability of rich clinical features, a few major issues may call into question some of the main arguments presented.

1. Endpoint definition: The PRS used are for coronary heart disease, the analyses works with a myocardial infarction endpoint. While closely related, this likely prevents an accurate representation of the predictive performance of the PRS used in this study, and likely underestimates it, especially since the models on the other feature sets are optimised for the exact endpoint definition (MI) they are also evaluated on. The use of PRS for CHD is even more puzzling since 8 PRS for MI exist in the PGS-catalogue (accessed via EFO_0000612)
2. LR and NN models perform virtually at the same performance as measured by ROC AUC, yet quite a few of the arguments rely on different precision/recall behaviour at the same decision thresholds which is likely due to just a different model calibration. It offers no insight about the model's ability to stratify between high- and low-risk individuals. The statement found across the manuscript that electronic health records provided much greater value-add for prediction, especially when used in a neural network, should be rephrased.
3. Please clarify train/test/valid splits and assure that presented performance is on entirely held out test splits (also not used for model selection). This part is confusing at the moment, especially since 70/30 splits are also mentioned in the manuscript. If models were evaluated on the model selection split their performance is likely overestimated.
4. Similarly, exact inclusion/exclusion criteria for the VAE and NN/LR models, as well as which EHR data was included from which timepoints and how they relate to the 10-year window, would be crucial to get an accurate impression of what information was available in the modelling setting.

Given these issues the main contribution seems to be, that while modelling 10-year MI risk based on EHR works better than based on genetics, a combination of both provides some additional improved performance. At the same time this seems to be a linear problem in which NNs do not provide a benefit for risk modelling (aside from the learned VAE representations that are outside the scope of this work). The reported feature importances match expectations from the literature around cardiovascular risk factors, and are independent of whether PRS are included in the model or not.

General comments:

Following points would be helpful to increase the strength of this study:

1. A PCA or no dimensionality reduction instead of the VAE-representations for the NG2 feature set would be highly interesting, since it would simplify obtaining the NG2 feature vectors to a simple matrix multiplication, thus being preferable if it reaches a similar performance. Then, the entire method would be linear (using the LR method) and more directly interpretable.
2. Age+Sex baselines e.g. in Fig 1c, 2a, 2c would be helpful as a rough lower bound for 'trivial' performance.

3. Investigating multiple endpoints covering different disease mechanisms would be a major improvement to the manuscript as it would allow for much broader claims (already implied by the title) than just regarding coronary artery disease, and given the setup of working with existing PRS from the PGS catalogue and the other features being virtually endpoint-agnostic, this would fit well with the presented approach.

4. Similarly, evaluating the presented models across ancestries in the UKB would be of interest, and could illustrate differences in ancestry-specificity between PRS and EHR, adding to the analysis of their differences.

Specific comments:

In quite a few instances, this reviewer would have expected information to be provided in a different section. Consider moving some of the details from Supplementary Methods to main text, readers have to know which PRS, incl. ancestry groups, datasets used for GWAS and PRS development, differences between PRS from just the main text

this should be: not based on UKB, or argued why it is OK to use UKB here, in a comparable ancestry to the UKB
S8 cite PGS, not just PGS catalogue and the IDs, use PRS ID for khera (PGS000013) instead of publication ID, which contains additional PGS on other diseases

S9 PGS000042 seems incorrect, it is for celiac disease, PGP000042 is for CAD

S9 'utilized data from online supplemental data for the other scores' which data, from which supplements?

S14 very hard to understand, I am not sure what an 'experiment', a 'trial', 'a random search of hyperparameters' are here.

S14 why are there 20 possible combinations of hyperparameters? Are values selected from a list of values, rather than continuously? If so, add a table with the lists of possible values?

Additional notes to be considered are:

2 Title leaves out what the models are trained on, should not focus on NNs when LRs performs on par

59 mention genotyping only has to be done once vs. repeatedly for non-genetic data

73 one could argue, that a 10-year risk prediction setting favors non-genetic-data based methods, while a 40-year risk prediction setting would favor a genetic-data-based method more. More clearly state timeline of interest and why 10 years?

79 would not call all of these NLP methods per se, though they originated in that context

89 cite existing work combining PRS + additional data in NNs, e.g. DeepNull for GWAS with NNs in the covariate modelling part

93 CAD is not the same as MI – whichever it is, please make text and figures consistent (see e.g. Fig 1c axis labels vs. figure caption)

96 crucial details: how many features in each category, roughly what is meant by 'diagnostic features derived from EHR'? diagnoses, correct?

96 so there are two NNs: one to produce feature set NG2, and then one for risk prediction? clarify, mention autoencoder in text, not just figure and methods

102 mention number of layers, number of parameters for better understanding of the models

704 label color bar in c, also to avoid confusion with y-axis label from the bar chart next to it

706 describe confidence intervals and how they are determined

706 Correlation of 0.12 between two PRS for the same phenotype might be concerning low, is there an explanation for this?

706 why not add purely non-genetic scores to correlation analysis? Would also give a better frame of reference for 0.12 correlation between CAD_1.7M and CAD_57

706 0.56 AUROC for the Khera PRS would be quite low for the European population (which this particular PRS is optimised for), one would expect closer to ~0.6 from others prior work using plink2 over the UKB imputed genotypes. Is this minor performance due to a difference in phenotype (CAD in the PRS development, MI used in this work)? Khera et al. specify their exact phenotype definition for CAD, which is a combination of multiple ICD9/ICD10/OPS-4 codes, incl. ones for myocardial infarction

103 clarify which PRS are being used before showing them in Figure 1c (like Sup. Tab. 6, but include original publications for reference). Also make their names more descriptive. How does 'All_CAD_PRSs' work exactly? at least mention if it is an ensemble, combination of weights, etc. in the main text

118 with deltaAUC this low, a net reclassification improvement index would be helpful to get a better understanding of the potential real-world effect of adding/removing the PRS

Fig. 2:

720 a) use same level of precision $0.0091 = 0.01 = 0.011$?

720 b) label b) missing

720 c) broken legend, not clear which is which

720 c) show datapoints, not density/fit, or at least add datapoints at every percentile / every fifth percentile, this way it is very hard to tell, what is actually happening at the 99th percentile (when comparing with Sup. Tab. 1, the raw data goes up to 17%, not sure a polynomial fit is a good way to model this)

720 d,e) use ROC plot instead: exact decision thresholds are not of interest, the achievable precision for a given recall and vice versa are. In these figures, NG2+PRS,NN (red) clearly trades in precision for recall compared to NG2+PRS,LR while according to fig2a) they have virtually the same ROC AUC, so I'm assuming the difference here is purely due to different calibration (score distributions in fig2f are also quite different, much more extreme for NG2+PRS,NN)

736 specificity you're referring to Sup. Tab. 1 here

S642 Table S1: what are the two numbers in each cell? Incidence rate, and what is in the ()?

Why are results in the left half not marked in bold when exceeding the 76.2% threshold?

Fig. 3:

752 Fig3 a) with just a handful PRS show individual datapoints instead of boxplot

752 Fig3 b) explain significance threshold used, incl. multiple testing adjustments

167 when discussing the NG2 features, please use Sup. Tab. 2 labels, also in the figure.

189 the argument about adding the PRS could be further strengthened by including other top-features in a correlation analysis like Fig. 1c), presumably showing relatively low correlation and further establishing the PRS independent nature
207 as discussed above, this reviewer is very sceptical about this recall argument given the near-identical ROC AUC values for NN and LR models

267 given the near-identical performance of NN and LR, this reviewer does not see features leveraged more effectively

277 typo \the

325 the clinical risk factors (e.g. Framingham) would still be available though

345 disagree, near-identical ROC AUC instead shows that the prediction problem can be solved with a linear model and NN does not capture any additional information

351 Were analyses only done in the white-british ancestry? Specify this much earlier than in the discussion, incl. how exactly the study population was defined, e.g. exact ICD-10 codes excluded

Methods section:

407 specify models were selected on the valid dataset, and reported performance is from the test dataset? If performance is reported on the same dataset on which models were selected it is likely overestimated. At other points e.g. S883 a 70/30 split is mentioned.

Unclear which cases are defined as prevalent. Should be those before assessment center 1 which the other features are from. Please specify which EHR data were used in the risk models, should also only be up to assessment center 1.

Please also discuss what data the VAE was trained on, and how this might impact this specific setting, e.g. was the VAE trained with MI / CHD diagnoses? Was it trained with EHR cut off at assessment center 1? From Supplements II23 it sounds like it was trained on the full dataset, perhaps mention this earlier in the text. It implies that the VAE-based vectors can already contain predictive information rather than just being a compressed representation of the patient history up to the start of the 10-year risk prediction window. This might be a concern if the train split of the VAE is included in the test-split of the NN/LR risk models, essentially leaking test-set data into the training process.

569 Code availability: UKB data cannot be made available by the authors, but code should be made publicly available without requiring a request, unless there is a good reason currently not mentioned

S343 this indicates that sex (and likely age) should have been included in the VAE model to prevent this sort of confounding from obscuring which features are actually of interest (similar to sex and genetic PCs in GWAS)

885 how was the standard error of the delta-AUC computed?

Reviewer #2

(Remarks to the Author)

This study evaluated models for myocardial infarction risk prediction in UK Biobank. In addition to previously reported polygenic risk scores (PRS), the authors used clinical information from reported coronary artery disease risk factors (NG1) and 574 variables representing medical history created from a complex representation learning algorithm (NG2). They used two different models: logistic regression and deep neural network. They show that the clinical data has better value for risk prediction than the PRS, although the best performing model includes both the clinical data and the PRS.

There rationale of the study is interesting and the methods seem sound, but the manuscript is sometimes hard to follow. Some analyses and their interpretation should be clarified and better put into perspective.

1) Please clarify the definition of the outcome (myocardial infarction) used in the study in the main text, including the diagnostic codes used. The number of cases/total individuals included in the analysis, including the number and justification for exclusion should also be added to the main text. The numbers given in Supplementary Table 5 do not seem to correspond to the White British individuals in UK Biobank. What does the "three diseases of interest" refer to (line 442)?

2) It seems like the authors considered only incident events, but defined a case group and a control group. Why not use Cox proportional hazards models?

3) The derivation of the diagnostic features (NG2) is complex and hard to interpret. The authors mention having validated the method in other datasets but no reference is provided (line 479). There seems to be a lot of overlap in the information contained in the 574 variables generated (e.g., same ICD10 code present in multiple variables), are they all independent or do some of them rely on similar information?

4) Including that many (correlated?) variables in a regression model is likely to cause collinearity. How does the training

method used account for this?

5) The features are likely to be very specific to the population / healthcare system in which they were developed. This could overestimate their predictive performance since they were trained from a large and relatively homogeneous database (UK Biobank). Please comment on the transferability of this method to different populations and settings.

6) The Results section is sometimes repetitive (lines 203-210, 238-243), I suggest to shorten and/or reorganize.

7) I do not understand the rationale for expecting a better added value of PRS to a model including more clinical information. Usually it becomes more difficult to improve the AUC of a model when good predictors are already in the model. Is there prior evidence to expect "potential effects of non-linear interactions between PRS and clinical features" (line 130)? I suggest to rephrase the title of the first section in the Results (lines 107-108).

8) Were the PRS correlated with the features included in NG2? It would help to determine if some genetic information were captured from the medical history and learning algorithm.

9) I suggest adding more information in the Discussion on the interpretability of the diagnostic features. Some of the elements in the Supplementary Discussion could be summarized in the main text. It seems for example that the top diagnostic features captured elements related to aging, sex, obesity smoking and lipids.

Minor points

1) Please use the term myocardial infarction consistently ("heart attack" is a lay term).

2) Please list the risk factors included in NG1 in the main text (Supplementary Table 7).

Reviewer #3

(Remarks to the Author)

In this manuscript, the authors report the results of a study based on UK Biobank data aimed at developing a myocardial infarction risk prediction score. The novel aspect is addition of polygenic risk scores to algorithms that include either a set of traditional cardiovascular risk factors or a broad array of information obtained from electronic health records. The authors conclude that addition of a polygenic risk score (PRS) to predictive models that are based on demographic and clinical data only minimally adds to risk prediction, independent of the modeling strategy.

1. The manuscript is well written and provides extensive data in support of the final conclusions. However, the use of the area under the curve (AUC) as the only model diagnostic criterion has its limitations. Although, there is admittedly no perfect way of deriving risk prediction models, the authors may want to consider examining model calibration and net reclassification, in addition to the AUC.

2. A flow chart showing inclusion/exclusion criteria would have been helpful. This chart should also indicate loss to follow-up at 10 years.

3. Use of the ICD10 codes in the development of the risk prediction models underutilizes the potential of EHRs. Provider notes can be messy and natural language processing protocols are not straightforward, however information contained in those notes can go significantly beyond diagnostic codes, which vary greatly in performance.

4. It would have been helpful if the authors differentiated between EHR for different types of services (ambulatory care, hospitalizations, etc.). This would have added to the clinical utility of the EHR-based scores.

5. I was not convinced by the rationale to use multiple PRSs in one model. Although considered a priori, this further approach complicates interpretation of an already complex analysis. Separate analyses with the different PRSs would have been more transparent and justifiable.

Version 1:

Reviewer comments:

Reviewer #2

(Remarks to the Author)

The authors addressed most of my comments adequately. I have a few remaining remarks:

1) Some confusion remains regarding the number of individuals included and the exclusion criteria. The authors now indicate:

<i>The 10-year incident case and control counts were 10,918 and 455,775. A total of 35,948 individuals who did not self-

report as being of White British ancestry or who were lost to follow-up in the incident prediction time period, as per data field 191 and the death registry, were excluded from the analyses unless they had an incident diagnosis of the disease (Supplementary Table 5).

Does this mean that individuals of all ancestries were included if they had an incident myocardial infarction? Also, the total number of individuals when adding the prevalent cases and controls in Supplementary Table 5 seems to include all ancestries.

2) The selection of the outcome as myocardial infarction yes/no over a follow-up period of 10 years while excluding all events occurring before recruitment could favor predictors of late-onset events. PRS have been shown to better predict early-onset events, which might explain their lower performance in the current study. I suggest to discuss this point and/or add as a limitation.

3) Regarding the transferability of this method to different populations and settings, I still believe that the information/performance of the EHR could differ according to population-dependent factors such as healthcare organization, prevalence of disease and risk factors, even cultural differences among the same ethnic group (European ancestry in this case).

4) Regarding the correlation between PRS and EHR features, the statement "The correlations with PRSs were all near zero" is imprecise, I suggest to add the PRS to the variables reported in Supplementary Figure 8 to support it.

Reviewer #3

(Remarks to the Author)

I appreciate the exhaustive responses provided by the authors to the initial critique. However, several remaining aspects need to be addressed.

1. The overall premise of this article is that PRS provide added value to myocardial infarction (MI) risk prediction models. However, study results suggests that this value is minimal. Additionally, the stated utility of PRS in clinical settings is impractical, at best. Therefore, contrary to the authors' concluding statements, study findings do not support an increasing role of PRS in clinical practice.

2. Please provide information on how do the results based on the NG1 model only compare to existing MI risk prediction models.

3. The authors may have mis-interpreted the initial request to consider as independent predictors in the expanded NG2 model, ICD codes for encounters that occurred in different clinical settings. In other words, there is clinical relevance to risk prediction from just ambulatory care data, in the absence of evidence of hospitalizations (within a defined time period). The same would be true for ICD codes obtained solely from Emergency Department visits.

Reviewer #4

(Remarks to the Author)

As an editorial board member at Nature Communications Medicine, I was asked to evaluate the rebuttal and revised manuscript of Hornbeck et al., entitled "Machine learning for myocardial infarction risk prediction: greater value-add from electronic health records than polygenic risk scores." This review was conducted in the absence of Reviewer #1, who is no longer available to assess the authors' responses and amendments.

In this study, the authors investigate and compare the predictive value of polygenic risk scores (PRSs), established cardiovascular risk scores, and extended electronic health record (EHR)-derived features using logistic regression models and feed-forward neural networks to predict 10-year myocardial infarction risk in the UK Biobank. Their findings indicate that PRSs contribute only a small but statistically significant improvement when combined with other readily available risk factors. This effect was more pronounced when PRSs were added to a limited clinical feature set rather than a more comprehensive EHR-based model. Additionally, the authors examined which machine learning methods were most effective, which features were most influential, and how these features interact within predictive models.

The review provided by Reviewer #1 was exceptionally thorough and of high scientific quality, reflecting a deep level of expertise. Based on my evaluation, I find that the authors have adequately addressed the reviewer's comments and concerns, making reasonable and well-justified amendments to the manuscript. In several instances, their revisions significantly improved the clarity and scientific rigor of the paper. Additionally, in cases where the authors chose not to implement certain suggested changes, such as the request to incorporate principal component analysis (PCA), I believe these were appropriate decisions. Adding PCA, for instance, would not have provided substantial additional value to the study but would have introduced an unnecessary layer of methodological complexity. I also found their rationale for using PRSs for coronary artery disease (CAD) rather than myocardial infarction (MI) to be well-argued. While this choice introduces a potential limitation, I believe the authors have adequately acknowledged and justified it in their latest revision. Overall, I believe the authors have satisfactorily addressed all of Reviewer #1's critiques, and I do not see the need for further modifications to the manuscript.

From my perspective, this amended manuscript is of exceptionally high scientific quality. The attention to detail, methodological rigor, depth of analysis, and data presentation demonstrated by the authors is extraordinary, particularly for a

machine learning-based study. In fact, their analytical approach—while impressive—was almost too exhaustive, to the point where the complexity of the methodology slightly detracts from the overarching message, making it less accessible to broader scientific and clinical audiences.

I also found the central conclusion of the manuscript particularly compelling and thought-provoking: adding polygenic risk scores (PRSs) resulted in only a marginal improvement in predictive performance for myocardial infarction risk compared to models incorporating clinical and EHR-derived data alone. This finding raises important questions about the practical utility of PRSs in cardiovascular risk assessment, suggesting that they may not provide meaningful prognostic insights beyond traditional risk factors or those that may soon be accessible through alternative approaches, such as natural language processing applied to big data. More critically, it challenges the notion that PRSs could play a significant role in altering disease trajectories or mitigating future cardiovascular events.

Final Assessment

The revised manuscript demonstrates scientific excellence, and the authors' responses to Reviewer #1 reflect thoughtful engagement with their critique. I believe the paper has improved and is acceptably ready for publication.

Version 2:

Reviewer comments:

Reviewer #2

(Remarks to the Author)

The authors have satisfactorily answered my remaining comments.

Reviewer #3

(Remarks to the Author)

The authors provided exhaustive responses to reviewer's comments. I have no further questions, but I would like to suggest that the authors consider deleting the last sentence of the abstract as it implies that the PRS scores provide added value to risk prediction, which is not the case in this study.

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Response to the Reviewers

General Response:

We thank the reviewers for the insightful comments that will enable us to improve the quality of our paper. We are pleased to re-submit the manuscript now titled “Machine learning for myocardial infarction risk prediction: greater value-add from electronic health records than polygenic risk scores” with corrections based on the reviewers’ comments. We have carefully considered each suggestion and have made corresponding updates to the manuscript.

The revised text is written in blue in the manuscript, and in this document we include excerpts of the **revised text (blue)** presented in the context of original **unchanged text (gray)** for ease of reading and understanding the textual context in which the changes were made. The reviewer comments are included in **orange** in this document for ease of visualization. We present our responses point-by-point based on the reviewer comments. We hope that this response letter and our updates to the manuscript will effectively address the questions and suggestions raised by the reviewers and enable our paper to be ready for publication.

Referee expertise:

Referee #1: Neural networks, UK biobank, heart

Referee #2: polygenic risk scores, myocardial infarction

Referee #3: Electronic health records, myocardial infarction

Reviewer #1 (Remarks to the Author):

In this article, the authors investigate and compare the predictive value of polygenic risk scores, established cardiovascular risk scores and extended EHR-derived features in logistic regressions and feed-forward neural networks for 10-year risk of heart attacks in the UK-Biobank. More specifically, they find that while PRS are easily outperformed on their own by clinical and EHR-features across model types, they do add low but significant value in combination with them, and more so for the smaller clinical feature set than the larger EHR-based one. They further investigate which of these features are most important, and how they interact with each other.

While the authors address the highly relevant question of PRS utility in the light of different machine learning models and availability of rich clinical features, a few major issues may call into question some of the main arguments presented.

Q1. Endpoint definition: The PRS used are for coronary heart disease, the analyses works with a myocardial infarction endpoint. While closely related, this likely prevents an accurate representation of the predictive performance of the PRS used in this study, and likely underestimates it, especially since the models on the other feature sets are optimized for the exact endpoint definition (MI) they are also evaluated on. The use of PRS for CHD is even more puzzling since 8 PRS for MI exist in the PGS-catalog (accessed via EFO_0000612)

We thank the reviewer for pointing this out. The 8 PRSs currently available on PGS specifically for myocardial infarction (MI) were published collectively in three studies by Jung et al. (2024), Tanigawa et al. (2022), and Sinnott-Armstrong et al. (2021). While it would be optimal to use PRSs that were created specifically using MI as a disease endpoint, this was not possible in our study for the following reasons.

All three of the above studies used the UK Biobank (UKB) to design the published PRSs for MI. More specifically, similarly to our study, they also used the White British population subset. Tanigawa et al. and Sinott-Armstrong et al. studies both leveraged the batch screening iterative lasso (BASIL) approach on the snpnet package in R to derive their scores on a train set within the UKB White British population, and then tested their scores on a holdout set in the same population. For example, Tanigawa et al. separated their White British population dataset into an 80/20 train/test split, and Sinott-Armstrong et al. used 70% of the UKB White British population as a train set, using the rest as validation and test sets. Importantly, for each of those studies the PRS weights and SNPs were learned and selected, respectively, using a large proportion of what comprises our population dataset (i.e., White British) within the UK Biobank. This results in a significant risk posed to the validity of our results, attributed to potential overfitting, were we to use these scores.

To safely leverage these PRSs with minimal risk to the validity of our study, we would need to ensure that the train and validation set results within the UK Biobank, as published in the original studies, are commensurate and not significantly distinct. Unfortunately, we were unable to find this comparison data in the Jung et al. and Sinnott-Armstrong et al. papers or in their supplementary materials. We did find comparison results for the Tanigawa et al. study in the online supplement which confirm the presence of potential overfitting in their score within the UK Biobank White British population. The AUCs of the PRS-only model on train and test sets, respectively, were 0.735 and 0.586, and when they compared a model with covariates only (age, sex, and genotype principal components) to a model with PRSs added, the respective AUC improvements for the train and test sets were 0.049 and 0.011. This suggests possible overfitting in the train set. While this may not be an issue if we were to have used a completely independent dataset, including this PRS (and possibly the others) in our study, which is based on the UK Biobank dataset, would risk the validity of our results, as some participants might have artificially better-performing PRSs based on their membership in the train set of the study that created the PRSs.

The Jung et al. study used a more traditional approach to set up their PRS for myocardial infarction, which involved running genome-wide association study (GWAS) to establish the PRS weights, followed by LDpred to select the variants. This study also used the UKB White British population for both GWAS and LDpred, where they split the population in half, using each respective half for the two steps involved in creating the PRS. Again, given that they used 50% of our study's population to establish weights for GWAS, there is the potential risk of overfitting.

For these reasons, we could not use these myocardial infarction-specific PRSs in our study. The PRSs that we selected all leveraged weights derived from independent GWAS studies using large population-level consortia. For example, the weights from our CAD_202, CAD_6M, and CAD_1.7M scores were derived from CARDIOGRAMplusC4D / 1000Genomes large-scale GWAS analyses. The CAD_202 and CAD_57 variants were selected based on false discovery rate and p-value from the original GWAS, respectively, and thus not influenced by the dataset. For CAD_1.7M score, LD pruning was used to select SNPs at different linkage disequilibrium thresholds, which again was independent of the dataset used, even though the UK Biobank was used for evaluating the PRSs once generated. Finally, while the CAD_6M score used LDpred, which leverages information about the disease phenotype (in this case CAD), and was created using the UKB, the authors compared the prediction performance for the

population subsets used to run LDPred and to evaluate results, and found almost equivalent results. These reasons justified the use of the selected PRSs in our study.

While we concede that our study may potentially be stronger if we were to include PRSs designed specifically for MI, this was not a viable choice for our current study, given the reasons above. If we were to repeat this study on a separate population dataset other than UK Biobank, perhaps then we could more reasonably leverage those available scores. We reviewed the case definitions in the studies that published the PRSs that we used, and all studies include myocardial infarction as a subset of the CAD case definition. Several also included patients with a history of procedures such as percutaneous transluminal coronary angioplasty (PTCA) or coronary artery bypass grafting (CABG), which are often recommended to patients presenting with angina that have > 70% stenosis in at least one coronary artery or with acute myocardial infarction (100% stenosis). Given that the etiology of the disease is roughly the same, whether we look at the more general CAD definition or the more specific myocardial infarction event definition as a subset of CAD cases, we would hope that this lack of myocardial infarction-specific PRSs used in our study would not have a significant impact on the results. Furthermore, we emphasize that the other feature sets (NG1 and NG2) comprise general risk factors and EHR-based features, respectively, that are not derived or optimized specifically with bias to myocardial infarction as an endpoint, thus not having MI-specific PRSs in our study does not introduce a bias in terms of equivalency in this regard.

In light of this, to account for the reviewer's concern, we have updated the Discussion section to include a mention of this as a limitation of the study. We also added references to the studies that created the MI-specific scores on the PGS catalog.

Change made to the Discussion section:

Another limitation relates to the PRSs that were used. Rather than just one, we selected several PRSs to leverage genetic information that might be independently contributed by each score. However, we could have selected a broader swathe of PRSs available in the literature, **including scores designed specifically for myocardial infarction**, and this is warranted for future work. **The granularity of case phenotype definitions used in designing a PRS can have an impact on score results, so a future study leveraging MI-specific PRSs may be warranted.**

Q2. LR and NN models perform virtually at the same performance as measured by ROC AUC, yet quite a few of the arguments rely on different precision/recall behaviour at the same decision thresholds which is likely due to just a different model calibration. It offers no insight about the model's ability to stratify between high- and low-risk individuals. The statement found across the manuscript that electronic health records provided much greater value-add for prediction, especially when used in a neural network, should be rephrased.

Thank you for pointing this out. We have rephrased sections from the text.

Changes made to the Title:

We changed the title from "Neural networks for myocardial infarction risk prediction: greater value-add from electronic health records than polygenic risk scores" to "**Machine learning** for myocardial infarction risk prediction: greater value-add from electronic health records than polygenic risk scores", as logistic regression and neural networks are both machine learning algorithms and this would serve as a broader term to better encompass the scope of the study.

Changes made to the Abstract:

We found that PRSs provided minimal value-add for risk prediction independently, while large-scale diagnostic data from electronic health records (EHRs) provided much greater value-add for prediction. ~~especially when used in a neural network.~~

Q3. Please clarify train/test/valid splits and assure that presented performance is on entirely held out test splits (also not used for model selection). This part is confusing at the moment, especially since 70/30 splits are also mentioned in the manuscript. If models were evaluated on the model selection split their performance is likely overestimated.

We apologize for any confusion on this matter. For the primary analyses mentioned in the main text, the train/validation/test splits were indeed 70/20/10. As mentioned in **Experimental Design and Model Training**:

Prior to starting any experiments, the UK Biobank samples were randomly split into train, validation, and test sets ten times (ratio of 70/20/10), and each random split was used to train all models for the disease to prevent per-sample variability from having a confounding impact on the results. Random search-based hyperparameter tuning was done during training for each trial.

We added the following to the last sentence in the section:

After training, the mean and standard deviation of the performance metrics were computed for each model to get a confidence interval [using the test set](#).

In the **Supplementary Results, Individual Performance of PRSs and Diagnostic Features**, we used a 70/30 train/test split, controlling for class imbalance, to compare the baseline performance of each PRS score using a univariate logistic regression model. The purpose of this analysis was to analyze how each score performed in isolation, prior to making a decision to combine, as there was no need for a validation set using this method.

Q4. Similarly, exact inclusion/exclusion criteria for the VAE and NN/LR models, as well as which EHR data was included from which timepoints and how they relate to the 10-year window, would be crucial to get an accurate impression of what information was available in the modelling setting.

Given these issues the main contribution seems to be that while modelling 10-year MI risk based on EHR works better than based on genetics, a combination of both provides some additional improved performance. At the same time this seems to be a linear problem in which NNs do not provide a benefit for risk modelling (aside from the learned VAE representations that are outside the scope of this work). The reported feature importances match expectations from the literature around cardiovascular risk factors, and are independent of whether PRS are included in the model or not.

Overall, we agree with your assessment on the main contribution.

As for the VAE, as we discuss in **Electronic Health Record (EHR) Representation Learning (Derivation of Non-Genetic Feature Set 2; NG2)** in **Supplementary Methods**, there were many design decisions that went into the medical history construction of NG2. The UK Biobank includes EHR data from five sources – hospital and primary care, death registry, cancer registry, and self-reported questionnaires. Some of the datasets contained sub-datasets of diagnostic codes using various coding systems, such as ICD-9.

To maximize the scope of diagnostic data we could leverage for pretraining the VAE, we decided to use all five data sources. One reason was that only roughly 18% of UK Biobank participants had data in only the hospital dataset, and we wanted to maximize available data for each participant. We mapped ICD-9, read version 2 codes, clinical

terms version 3 codes, and a UK Biobank designed coding system, to the ICD-10 coding system. Next, after mapping the codes to ICD-10, the next step was to consolidate diagnoses for each patient based on the initial recorded date. Given that some data, such as self-reported questionnaire responses, was retrospective and recorded annually, the dataset was organized by diagnoses per year to facilitate the merging process. For instance, if a patient had records in three datasets, and two of those contained diagnoses from different time points within the same year, the combined diagnoses would represent the patient's medical conditions for that year. Following this merging process, approximately 33,000 patients had no recorded diagnoses across the five datasets.

We generated 19.1k-dimensional sparse vectors representing the medical history of those with recorded diagnoses across the five datasets, compressing them into a 574-dimensional latent representation containing semantic information about the relationships between the input codes. Given that each vector, rather than patient, was considered a sample, each sample is considered a medical history in general, rather than a patient specific medical history.

The details on the derivation of the NG2 feature set are described in the **Supplementary Methods** and in **Supplementary Figures 1-3**, and we are in the process of finalizing a manuscript for publication that will describe our methodology in extensive detail.

General comments:

Following points would be helpful to increase the strength of this study:

GC1. A PCA or no dimensionality reduction instead of the VAE-representations for the NG2 feature set would be highly interesting, since it would simplify obtaining the NG2 feature vectors to a simple matrix multiplication, thus being preferable if it reaches a similar performance. Then, the entire method would be linear (using the LR method) and more directly interpretable.

Thank you for your comment. While we agree that linear models are generally easier to interpret, we believe there is significant value to our approach of leveraging the VAE to derive the NG2 feature set. First, each original medical history feature set containing all diagnoses for each UK Biobank participant up to the point of their first assessment visit in the UK Biobank (i.e., baseline) contained ~19,154 features comprising ICD-10 codes at various hierarchical levels. The scope of the feature set alone poses a major challenge for training a logistic regression or neural network model for disease prediction on the limited set of participants used in our study (i.e., less than ~500k individuals), resulting in the risk of model overfitting and the need for significant regularization to combat this. While the diagnostic features could be evaluated individually as univariate models, this would then defeat the purpose of the study in exploring the value-add of PRSs over non-genetic data as a feature set comparison. For this reason alone, it makes sense for us to use a smaller or compressed feature set.

While PCA is an excellent model, we argue that using this instead of the neural network model would instead make it more challenging to interpret the significant features. Say that, for each participant, we compressed their medical history data into a low-dimensional embedding using PCA. The feature sets would be established based on the amount of variance explained, which would be difficult to interpret. Similarly, other linear matrix factorization approaches would require their own bespoke approaches to understand the underlying meaning of the features.

While we could have used a selected subset of ICD-10 codes (i.e., all three-character codes), we believe that this might reduce the scope of available information leveraged in the model, as many individuals only have their data coded as more granular codes, and would thus be missed in the features provided. In our VAE, we trained using a bespoke approach that enables the model to capture similarities between ICD-10 codes from an expanded sample

set (~3 million), essentially learning the concept of a “medical history” similarly to how a large language model can learn the concept of “language”.

We provide extensive details on the interpretation of these diagnostic features in the **Supplementary Methods** (“Interpretation of Diagnostic Features - Methodology”, “Feature Interactions Analysis - Methodology”), **Supplementary Results** (“Interpretation of Diagnostic Features - Additional Results”), **Supplementary Discussion**, **Supplementary Figs. 4-8**, **Supplementary Table 2**, and the **Supplementary Data**.

GC2. Age+Sex baselines e.g. in Fig 1c, 2a, 2c would be helpful as a rough lower bound for ‘trivial’ performance.

While we agree with this comment, we feel that Figs. 1 and 2, including 1c, 2a, and 2c, are already highly crowded figures and would prefer not to distract the readers with additional information given the already complex details. For now, we have not made changes, however if the reviewer strongly suggests the added benefit of including these additional components to the figures, we can include them either on Figs. 1 and 2, or as supplementary figures.

GC3. Investigating multiple endpoints covering different disease mechanisms would be a major improvement to the manuscript as it would allow for much broader claims (already implied by the title) than just regarding coronary artery disease, and given the setup of working with existing PRS from the PGS catalog and the other features being virtually endpoint-agnostic, this would fit well with the presented approach.

We completely agree with this point. We do hope to expand to other diseases in future work, but primarily wanted to focus on the core of the analysis process with a well-studied disease. We do have plans to expand in the future, and are working on studying the model architectures in-depth from a pure machine learning perspective currently.

GC4. Similarly, evaluating the presented models across ancestries in the UKB would be of interest, and could illustrate differences in ancestry-specify between PRS and EHR, adding to the analysis of their differences.

We do agree with this. One of the main limitations, as we point out in the **Discussion**, is that we were (and currently still are) limited to individuals of self-reported White British ancestry, given they comprise the vast majority of data available to us in the UK Biobank. We are hoping to get additional datasets with more diverse ethnic backgrounds in the future, but do want to emphasize our general methodology could still be applied.

Specific

comments:

SC - In quite a few instances, this reviewer would have expected information to be provided in a different section. Consider moving some of the details from Supplementary Methods to main text, readers have to know which PRS, incl. ancestry groups, datasets used for GWAS and PRS development, differences between PRS from just the main text

Thank you for your feedback.

We have added additional information to **Cohort Definitions** related to ICD-10 selection.

We adapted mappings from each coding system to ICD-10 (Supplementary Table 3) and then selected ICD-10 codes that would reasonably identify individuals as being cases for myocardial infarction, I21 (I21.0, I21.1, I21.2, I21.3, I21.4, I21.9, I21.X), I22 (I22.0, I22.1, I22.8, I22.9), I23 (I23.0, I23.1, I23.2, I23.3, I23.4, I23.5, I23.6, I23.8), I24.1, and I25.2 (Supplementary Table 4).

We also added the bulk of the **Calculation and Selection of Disease-Specific Polygenic Risk Scores (PRSs)** section in **Supplementary Methods** to the **Preparation of PRS Features Set**, as we agree this is a more appropriate place.

We utilized data from chromosomes 1-22 to calculate each PRS for each individual. We downloaded the variant IDs, effect alleles, and weights (betas) from the Polygenic Score (PGS) Catalogue⁶¹ for CAD_6M (publication PGP00006, PRS ID PGS000013) and CAD_57 (publication PGP000042, PRS ID PGS 000057) and utilized data from online supplemental data for the other scores. Specifically, for CAD_1.7M, we used PGS00018 from study PGP000007 and for CAD_202, these SNPs and their weights (log odds) were derived from the CARDIoGRAMplusC4D 1000 Genomes meta-analysis⁶² by identifying independent SNPs at false discovery rate (FDR) < 0.05. Each score was checked to ensure the SNPs were in GRCh37 (hg19), to map to the UK Biobank dataset.

We also made sure to include the ancestry information in the **Introduction**.

To avoid potential ancestry-related confounders when analyzing the PRSs, our analysis centered on self-reported White British individuals only using UK Biobank datafield 21000. The PRSs used contained 57, 202, ~1.7 million, and ~6 million variants, respectively, and for the purposes of this study were named CAD_5746, CAD_20226, CAD_1.7M26, and CAD_6M11.

SC - S8 cite PGS, not just PGS catalogue and the IDs, use PRS ID for khera (PGS000013) instead of publication ID, which contains additional PGS on other diseases

The PRS ID for Khera has been added, thank you for pointing that out.

SC - S9 PGS000042 seems incorrect, it is for celiac disease, PGP000042 is for CAD

Fixed.

SC - S9 'utilized data from online supplemental data for the other scores' → which data, from which supplements?

Thank you for pointing this out.

We have added this information to the **Calculation and Selection of Disease-Specific Polygenic Risk Scores (PRSs)** section in **Supplementary Methods**.

CAD_1.7M: Used PGS00018 from study PGP000007 (Inouye et al. 2018) on PGS Catalog.

CAD_202: These SNPs and their weights (log odds) were derived from the CARDIoGRAMplusC4D 1000 Genomes meta-analysis (Nikpay et al.) by identifying independent SNPs at false discovery rate (FDR) < 0.05.

Specifically, for CAD_1.7M, we used PGS00018 from study PGP000007⁵ and for CAD_202, these SNPs and their weights (log odds) were derived from the CARDIoGRAMplusC4D 1000 Genomes meta-analysis⁶ by

identifying independent SNPs at false discovery rate (FDR) < 0.05. Each score was checked to ensure the SNPs were in GRCh37 (hg19), to map to the UK Biobank dataset.

SC - S14 very hard to understand, I am not sure what an ‘experiment’, a ‘trial’, ‘a random search of hyperparameters’ are here.

Sorry for the confusion. We agree this was not very articulate at first.

We changed the language in **Model Training and Selection Details** to:

For each experiment (e.g., NG1; NN, NG1+PRS) 10 trials (iterations with different train/validation/test samples) were run systematically, each comprising a random set of hyperparameters.

SC - S14 why are there 20 possible combinations of hyperparameters? Are values selected from a list of values, rather than continuously? If so, add a table with the lists of possible values?

Sorry for the confusion here.

Yes, you are correct. We have added the details to **Supplementary Table 9: Hyperparameter ranges for model training**, for both models.

Additional notes to be considered are :

AN - 2 Title leaves out what the models are trained on, should not focus on NNs when LRs performs on par

Agreed. Updated the title to **Machine learning for myocardial infarction risk prediction: greater value-add from electronic health records than polygenic risk scores.**

AN - 59 mention genotyping only has to be done once vs. repeatedly for non-genetic data

Agreed. This sentence was corrected.

AN - 73 one could argue, that a 10-year risk prediction setting favors non-genetic-data based methods, while a 40-year risk prediction setting would favor a genetic-data-based method more → More clearly state timeline of interest and why 10 years?

While this could potentially be a viable argument, it depends on how the question is framed. Hypothetically, say that a PRS were to be calculated very early in life between ages 0-20, prior to the onset of most non-genetic risk factors, to predict incident myocardial infarction risk 40 years later. In this case, as little or no non-genetic data are available to otherwise enhance baseline risk prediction, this PRS might be the best predictor available (AUCs ranging up to 0.6). However, if taken later in life, say around the ages of 45-65, the baseline non-genetic models would already be robust given the available data, and thus the value-add of a PRS would be similarly small (as per our analyses), whether for 10-year (or less) or 40-year incident risk prediction.

The rationale for using a 10-year predictor was based on the timeline of available data in the UK Biobank study. Participants were first recruited between 2006 and 2010, and at the time our analysis was conducted, our department only had access to hospital records data through 2018-2019. To maximize the number of incident

cases available in our dataset, we selected a 10-year follow-up time. We independently conducted analyses for 2-5 year follow-ups but decided against pursuing those further due to the small resulting sample size.

AN - 79 would not call all of these NLP methods per se, though they originated in that context

Agreed. Added the word related, changing to [natural language processing related](#).

AN - 89 cite existing work combining PRS + additional data in NNs, e.g. DeepNull for GWAS with NNs in the covariate modelling part

Thank you for pointing this out. We agree and have cited.

AN - 93 CAD is not the same as MI – whichever it is, please make text and figures consistent (see e.g. Fig 1c axis labels vs. figure caption)

Thank you for pointing this out. Please see our response to your first main question about endpoints. We have made modifications throughout the text with respect to removing “heart attack”. For Figure 1c we would like to leave this as is, as it refers to the nomenclature specific to each PRS (i.e., we referred to the PRSs as CAD_202, CAD_6M, etc., as they were derived with coronary artery disease as endpoint).

AN - 96 crucial details: how many features in each category, roughly what is meant by ‘diagnostic features derived from EHR’? → diagnoses, correct?

Thank you for pointing this out. The feature set for NG1 is described in detail in **Supplemental Table 7**. However, we acknowledge that it makes sense to make this clear for the readers early in the text. Thus, we added a note in this Introduction section that indicates that NG1 has 15 features and that NG2 has 574 features. Furthermore, we also included the specific features for NG1 in the **Methods** section, as previously they were only written in **Supplementary Methods**.

We used two categories of clinical (non-genetic; NG) feature space: a) a small set of [15 known established risk factor](#) data such as blood pressure, smoking status, and cholesterol that are often measured as part of existing clinical risk scores such as the FRS or ACC/AHA scores (non-genetic feature set 1; NG1), and b) a large-scale set of [574 diagnostic features](#) derived from EHRs using a deep representation learning algorithm (non-genetic feature set 2; NG2).

The "diagnostic features derived from EHRs" are detailed in depth in the **Methods** ("Derivation of NG2 Feature Set"), in the **Supplementary Methods**, in **Supplementary Figs. 4-8**, in the **Supplementary Results**, and in **Supplementary Table 8**.

AN - 96 so there are two NNs: one to produce feature set NG2, and then one for risk prediction? → clarify, mention autoencoder in text, not just figure and methods

Sorry for the confusion about this. NG2 is derived using an autoencoder framework (which is a neural network, to your point), which we described here in the **Introduction** as a representation learning algorithm. We wanted to be consistent with how we introduced the derivation, using a simple mention, with the description of neural network and logistic regression models in this section.

However, to be clear, when we refer to NN in the manuscript, we are referring to the neural network model for disease prediction. This is another reason we wanted to keep the nomenclature of the VAE used to derive NG2 as a deep representation learning algorithm, as to not confuse readers with variations of NN.

AN - 102 mention number of layers, number of parameters for better understanding of the models

Thank you for pointing this out.

We added this sentence to point to the new table we added based on another comment.

The experimental design framework is visualized in **Fig. 1** (see **Supplementary Table 9 for additional details on the hyperparameter ranges for each model**), and the representation learning algorithm used to derive the large-scale diagnostic features is described in the **Supplementary Methods** and in **Supplementary Figs. 1 and 2**.

AN - 704 label color bar in c, also to avoid confusion with y-axis label from the barchart next to it

Fixed.

AN - 706 describe confidence intervals and how they are determined

Thank you for pointing this out. We added the sentence:

After training, the mean and standard deviation of the performance metrics were computed for each model across 10 trials, for each test set prevalence at each percentile to get a 95% confidence interval using the test set.

To the legend in Fig. 2b to clarify this.

AN - 706 Correlation of 0.12 between two PRS for the same phenotype might be concerningly low, is there an explanation for this?

We find this to be an interesting phenomenon, and it is likely due to differences in the set of SNPs in each score as well as the respective weights, i.e., the scores are each capturing at least partially non-overlapping components of the heritability of MI/CAD. As a result of the lower-than-optimal correlations between the selected PRSs, we decided to use all of them combined as the PRS feature set, allowing a logistic regression or neural network model to then learn the optimal weighting for each score to derive the best predictions.

AN - 706 why not add purely non-genetic scores to correlation analysis? Would also give a better frame of reference for 0.12 correlation between CAD_1.7M and CAD_57

This is a great question. For Fig. 1, we primarily wanted to convey the derivation of features spaces for NG1 and NG2 and model complexity, so the correlation analysis in Fig. 1c was simply added to provide further depth to the understanding of the PRSs used in the study.

The rationale for showing the correlations for the PRSs here, along with the univariate scores for myocardial infarction risk prediction, was to show that if we used multiple PRSs collectively in the analysis they could capture slightly different independent components of genetic risk.

That said, we appreciate the suggestion. We have analyzed correlations between some non-genetic features in **Supplementary Figs. 7 and 8**, with a focus on the top NG2 features and some of the NG1 features. If the reviewer emphasizes the importance of this, we can further add an additional figure in the supplement that evaluates the pairwise correlations between all NG1 features.

AN - 706 0.56 AUROC for the Khera PRS would be quite low for the European population (which this particular PRS is optimised for), one would expect closer to ~0.6 from others prior work using plink2 over the UKB imputed genotypes. Is this minor performance due to a difference in phenotype (CAD in the PRS development, MI used in this work)? Khera et al. specify their exact phenotype definition for CAD, which is a combination of multiple ICD9/ICD10/OPS-4 codes, incl. ones for myocardial infarction

Thank you for pointing this out. In our previous study (Isgut et al. 2021, *Genome Medicine*), we also evaluated the performance of the Khera et al. PRS for CAD (using a similar phenotype definition) and for MI (using a similar phenotype definition to that in our current study). For CAD, we achieved a test set AUC of ~0.60, and interestingly, for MI we achieved an even higher performance of 0.63 with the score (results in **Supplemental Table 8** of the Isgut et al. 2021 study). This was without including other covariates in the model. The key difference between our previous work and the Khera et al. study in comparison to our current work is that we evaluated 10-year incident risk, and thus we excluded any cases with their initial diagnosis prior to assessment visit 1 as well as controls lost to follow-up within the time window. This difference reduced the overall case count and is a possible explanation for the lower score.

AN - 103 clarify which PRS are being used before showing them in Figure 1c (like Sup. Tab. 6, but include original publications for reference). Also make their names more descriptive. How does 'All_CAD_PRSs' work exactly? at least mention if it is an ensemble, combination of weights, etc. in the main text

We appreciate the feedback. We added this sentence to the fourth paragraph in the Introduction, so they are referenced before Figure 1.

The PRSs used contained 57, 202, ~1.7 million, and ~6 million variants, respectively, and for the purposes of this study were named CAD_5754, CAD_20226, CAD_1.7M26, and CAD_6M11.

All_CAD_PRSs is a combination of all PRSs (four features were thus used in the logistic regression model used to calculate the Mean Test AUC in Figure 1 C, right), and is only used for Figure 1c.

AN - 118 with deltaAUC this low, a net reclassification improvement index would be helpful to get a better understanding of the potential real-world effect of adding/removing the PRS

We do agree that NRI is an emerging technique that is used as an alternative to AUC, and while we have used it in some other work, we were hesitant to do so in this paper. One reason is that AUC is still the de facto measure in most machine learning papers. Also, work by Pepe et al. has shown that NRI can be positive even when additional features add no predictive information when using large datasets, such as the one we have.

Please let us know if this is satisfactory. If you disagree, we can certainly look into adding it to the analysis.

AN - Fig. 2:

AN - 720 a) use same level of precision $0.0091 = 0.01 = 0.011$?

Fixed.

AN - 720 b) label b) missing

Fixed.

AN - 720 c) broken legend, not clear which is which

Fixed.

AN - 720 c) show datapoints, not density/fit, or at least add datapoints at every percentile / every fifth percentile, this way it is very hard to tell, what is actually happening at the 99th percentile (when comparing with Sup. Tab. 1, the raw data goes up to 17%, not sure a polynomial fit is a good way to model this)

Fixed.

AN - 720 d,e) use ROC plot instead: exact decision thresholds are not of interest, the achievable precision for a given recall and vice versa are. In these figures, NG2+PRS,NN (red) clearly trades in precision for recall compared to NG2+PRS,LR while according to fig2a) they have virtually the same ROC AUC, so I'm assuming the difference here is purely due to different calibration (score distributions in fig2f are also quite different, much more extreme for NG2+PRS,NN)

We appreciate this suggestion. We think these Figures 2d and 2e are nonetheless interesting, but agree with the reviewer on the relevance of using the ROC plot. We have added this to the Supplement as **Supplemental Figure 9**.

AN - 736 specify you're referring to Sup. Tab. 1 here

Fixed.

The incidence at each percentile of risk is shown for all the logistic regression models (see **Supplementary Table 1 for more details**).

AN - S642 Table S1: what are the two numbers in each cell? Incidence rate, and what is in the ()?

Sorry about this, this is the standard error. Fixed.

AN -Why are results in the left half not marked in bold when exceeding the 76.2% threshold?

Thank you for asking this. These values are not on the fitted curve, so we did not bold them.

AN - Fig. 3:

AN - 752 Fig3 a) with just a handful PRS show individual datapoints instead of boxplot

Fixed.

AN - 752 Fig3 b) explain significance threshold used, incl. multiple testing adjustments

This has been corrected. We added the following text to Fig3 b):

Significant features were selected after correcting for multiple comparisons based on the total number of features in each model using the Bonferroni correction ($\alpha = 0.01$).

AN - 167 when discussing the NG2 features, please use Sup. Tab. 2 labels, also in the figure.

We appreciate what the reviewer is trying to convey in that there is a lack of clarity upon first reading the paper on what diagnostic feature 235 is. However, we explain in depth within the next paragraph what the meaning is, and feel that including these ICD-10 codes in parentheses immediately after first mention of diagnostic feature 235 can be confusing to the reader. Given this, we have decided to leave this as is, in the hope that the reviewer agrees that the following paragraph fully explains what the feature means and properly guides the reader to the supplementary material for a more in depth understanding.

AN - 189 the argument about adding the PRS could be further strengthened by including other top-features in a correlation analysis like Fig. 1c), presumably showing relatively low correlation and further establishing the PRS independent nature

We agree with this point completely. We wanted to address the PRS correlations in the figures, and convey that other possible correlations in the models themselves were accounted for using regularization techniques such as L1, as we felt this conveyed the most valuable information visually without creating abnormally large figures. Please let us know if you agree with our reasoning.

AN - 207 as discussed above, this reviewer is very sceptical about this recall argument given the near-identical ROC AUC values for NN and LR models

We completely understand the skepticism here.

We feel the argument is valid. What we are observing is that the NN model is better at identifying positive cases (the minority class), albeit at the expense of precision (i.e., it is getting the positive classes correct, but introducing more false positives as well). While the trade-off between precision and recall is important overall, reducing false negatives in the case of disease incidence prediction is something we view as favorable, even in the case of having more false positives, as missing a diagnosis is more costly than a false positive.

AN - 267 given the near-identical performance of NN and LR, this reviewer does not see features leveraged more effectively

Thank you for pointing this out. We agree and changed the sentence to:

This suggests that use of a neural network does not change the trend of which features are most important, but rather leverages certain diagnostic features differently for prediction.

AN - 277 typo \the

Fixed.

AN - 325 the clinical risk factors (e.g. Framingham) would still be available though

We agree, and this is a great find. We removed “only-and” from the sentence to make it:

In these cases, PRSs combined with basic demographic information might be the best-available predictors, providing myocardial infarction risk prediction AUCs of ~0.6 alone and even higher when combined with demographic data.

AN - 345 disagree, near-identical ROC AUC instead shows that the prediction problem can be solved with a linear model and NN does not capture any additional information

Thank you for the feedback. We agree the word “useful” isn’t correct here.

We changed the sentence to:

These findings suggest that neural networks may be capturing ~~useful~~ non-linear trends between input diagnostic features, in line with the fact that neural networks can capture non-linear information that linear models cannot.

AN - 351 Were analyses only done in the white-british ancestry? Specify this much earlier than in the discussion, incl. how exactly the study population was defined, e.g. exact ICD-10 codes excluded

We agree and thank you for your feedback.

We have added the following information to the 4th paragraph of the **Introduction**:

To avoid potential ancestry-related confounders when analyzing the PRSs, our analysis centered on self-reported White British individuals only using UK Biobank datafield 21000.

Methods section:

M - 407 specify models were selected on the valid dataset, and reported performance is from the test dataset? If performance is reported on the same dataset on which models were selected it is likely overestimated. At other points e.g. S883 a 70/30 split is mentioned.

Thank you for pointing this out.

To clarify, yes, performance is reported on the test set, so that the model selected from the validation set is being tested on unseen data. We added a sentence to the end of the **Experimental Design and Model Training** section to clarify that performance metrics were on the test set. To be clear, the performance is not reported on the same data on which models were selected.

The points with the 70/30 train/test split are for univariate analysis and discussed in the **Supplementary Methods**. **Here**, we did use a 70/30 train/test split, controlling for class imbalance, to compare the baseline performance of each PRS score using a univariate logistic regression model. The purpose of this analysis was to analyze how each score performed in isolation, prior to making a decision to combine, as there was no need for a validation set using this method.

M - Unclear which cases are defined as prevalent. Should be those before assessment center 1 which the other features are from. Please specify which EHR data were used in the risk models, should also only be up to assessment center 1.

We apologize for the confusion. In the **Cohort Definitions** section, we made sure to include the following:

Any individual diagnosed with at least one code in the list at any point in time in their medical history would be considered as a prevalent case.

Thus, a prevalent case is any individual

M - Please also discuss what data the VAE was trained on, and how this might impact this specific setting, e.g. was the VAE trained with MI / CHD diagnoses?

Was it trained with EHR cut off at assessment center 1?

From Supplements ll23 it sounds like it was trained on the full dataset, perhaps mention this earlier in the text. It implies that the VAE-based vectors can already contain predictive information rather than just being a compressed representation of the patient history up to the start of the 10-year risk prediction window. This might be a concern if the train split of the VAE is included in the test-split of the NN/LR risk models, essentially leaking test-set data into the training process.

Thank you for bringing up this point. When deriving the NG2 feature set using the VAE, we were extremely careful to design the model to be timeline-consistent. The data used in the NG2 feature set comprised each UK Biobank participant's entire set of diagnoses in their medical history up until that individual's baseline (assessment visit 1), which maintains the consistency of the timeline with the other phenotypic data collected and with the 10-year incidence timeline. This larger dataset was then compressed using the pretrained VAE into a 574-dimensional embedding, as described in the paper.

What follows this is the question of whether the VAE itself leverages samples from a given individual from later points in time, while it is being trained. We concede that the training and data setup of the VAE was done using a bespoke approach that we describe as "patient-agnostic" - similarly to how a large language model can learn from a body of text from various sources and simply learns a representation of language itself, with this work we aimed to have the model learn a representation of commonly coexisting morbidities without prior knowledge of the time trajectory of any individual patient's set of diagnoses. Large language models trained to represent text, in essence, have learned conceptual patterns in language, and this concept can similarly be adapted to the idea of learning from clinical diagnostic data. In our study, each sample provided to the model was a randomly selected set of comorbidities that existed at the same point in time for any given patient (with the time and patient completely unknown and not represented in the model).

For this reason, we consider the NG2 feature set as a conceptual embedding not influenced by the diagnostic trajectory over time of any specific patient, and so we are not concerned about a data leakage issue. The NG2 feature set derivation will be pending publication from our group.

M - 569 Code availability: UKB data cannot be made available by the authors, but code should be made publicly available without requiring a request, unless there is a good reason currently not mentioned

The code for the analysis can be provided and made publicly available with the exception of the code used to generate NG2. For the code used to generate the NG2 features, we can make this available upon reasonable request, as we are planning on submitting another paper completely focused on this method and would prefer to wait until it is published to release that part of the code.

M - S343 this indicates that sex (and likely age) should have been included in the VAE model to prevent this sort of confounding from obscuring which features are actually of interest (similar to sex and genetic PCs in GWAS)

While we acknowledge the linkages to sex and age as discussed in the **Supplementary Discussion**, they were not of the proper data type to be included in the VAE model. The VAE model ingested vectors of diagnostic codes that were designed to be patient agnostic. Including such demographic information would have made the individual vectors patient specific.

M - 885 how was the standard error of the delta-AUC computed?

Sorry for the confusion here.

For the 10 trials in each model, the individual delta-AUCs were used. Thus, we calculated the standard deviation of the 10, and divided by the square root of the sample size.

Reviewer #2 (Remarks to the Author):

This study evaluated models for myocardial infarction risk prediction in UK Biobank. In addition to previously reported polygenic risk scores (PRS), the authors used clinical information from reported coronary artery disease risk factors (NG1) and 574 variables representing medical history created from a complex representation learning algorithm (NG2). They used two different models: logistic regression and deep neural network. They show that the clinical data and the PRS. There rationale of the study is interesting and the methods seem sound, but the manuscript is sometimes hard to follow. Some analyses and their interpretation should be clarified and better put into perspective.

Q1) Please clarify the definition of the outcome (myocardial infarction) used in the study in the main text, including the diagnostic codes used. The number of cases/total individuals included in the analysis, including the number and justification for exclusion should also be added to the main text. The numbers given in Supplementary Table 5 do not seem to correspond to the White British individuals in UK Biobank. What does the “three diseases of interest” refer to (line 442)?

Thank you for your feedback.

Please clarify the definition of the outcome (myocardial infarction) used in the study in the main text, including the diagnostic codes used.

We added this information directly to the **Cohort Definition** section in **Methods**.

We adapted mappings from each coding system to ICD-10 (Supplementary Table 3) and then selected ICD-10 codes that would reasonably identify individuals as being cases for myocardial infarction, I21 (I21.0, I21.1, I21.2, I21.3, I21.4, I21.9, I21.X), I22 (I22.0, I22.1, I22.8, I22.9), I23 (I23.0, I23.1, I23.2, I23.3, I23.4, I23.5, I23.6, I23.8), I24.1, and I25.2 (Supplementary Table 4).

The number of cases/total individuals included in the analysis, including the number and justification for exclusion should also be added to the main text.

Thank you for pointing this out. We altered the last sentence in **Cohort Definitions** to convey this information.

The 10-year incident case and control counts were 10,918 and 455,775. A total of 35,948 individuals who did not self-report as being of White British ancestry or who were lost to follow-up in the incident prediction time period, as per data field 191 and the death registry, were excluded from the analyses unless they had an incident diagnosis of the disease (Supplementary Table 5).

What does the “three diseases of interest” refer to (line 442)?

We apologize for the three diseases of interest in the text. That was a typo we made merging Word files. It has been removed.

Q2) It seems like the authors considered only incident events, but defined a case group and a control group. Why not use Cox proportional hazards models?

This is a good question.

The reason we did not use Cox proportional hazards models is that we were not trying to analyze the time until an event occurred. In our problem, we were addressing a binary classification (i.e., whether an event did occur or not), and the exact timing was not as relevant, just that it occurred during the 10-year window.

Q3) The derivation of the diagnostic features (NG2) is complex and hard to interpret. The authors mention having validated the method in other datasets but no reference is provided (line 479). There seems to be a lot of overlap in the information contained in the 574 variables generated (e.g., same ICD10 code present in multiple variables), are they all independent or do some of them rely on similar information?

Deriving the NG2 feature set using the VAE, we designed the model to be timeline-consistent. The data used in the NG2 feature set comprised each UK Biobank participant's entire set of diagnoses in their medical history up until that individual's baseline (assessment visit 1), which maintains the consistency of the timeline with the other phenotypic data collected and with the 10-year incidence timeline. This larger dataset was then compressed using the pretrained VAE into a 574-dimensional embedding, as described in the paper. We removed the text about other datasets, that was an error.

Q4) Including that many (correlated?) variables in a regression model is likely to cause collinearity. How does the training method used account for this?

We do acknowledge that this is a common problem using linear models.

To address potential collinearity problems, we used L1 regularization (also known as Lasso – Least Absolute Shrinkage and Selection Operator) in both the logistic regression and neural network models. For the neural network, we applied L1 regularization to both the input and hidden layers. This helped to shrink the coefficient of less important predictive features to zero, and handle multicollinearity by selecting only one variable from a group of correlated variables and shrinking the others to zero.

Q5) The features are likely to be very specific to the population / healthcare system in which they were developed. This could overestimate their predictive performance since they were trained from a large and relatively homogeneous database (UK Biobank). Please comment on the transferability of this method to different populations and settings.

Thank you for pointing this out.

We do acknowledge that a limitation of our study is that our analysis focuses on individuals of European ancestry alone, given that they are the most highly represented group in the UK Biobank, as mentioned in the **Discussion** section.

The method itself, however, is not hindered by the specific population. Given enough data, our methodology could be used to analyze any other population, assuming that the non-genetic factors used to derive NG1 and similar EHR data used to derived NG2 were available.

We hope to conduct future studies with various ancestries, and do point out that future studies should evaluate this question in the context of different ancestry groups to confirm whether the results are generalizable in the **Discussion** section as well.

Q6) The Results section is sometimes repetitive (lines 203-210, 238-243), I suggest to shorten and/or reorganize.

Thank you for pointing this out. We completely agree. Starting with the previous line 203, the beginning of There are interactions between PRSs and clinical features, but most of the value-add of PRSs is through their independent additive effect, we changed the text to:

While we found that independently increasing the model complexity had no significant impact on the PRS value-add as measured by AUC, the multi-modal NG1+PRS; NN model achieved significantly higher recall than the NG1; NN model at class decision thresholds 0.6, 0.7, and 0.8, a result that was not found in the respective logistic regression model comparison, suggesting that inclusion of PRSs in a neural network may improve model sensitivity for identifying at-risk individuals (Fig. 2 and Table 2).

To reflect the key point that the NN model achieved a higher recall, without reiterating previous results.

Similarly, for the text previously starting on line 238, the beginning of Multi-modality neural network models with large-scale diagnostic data, established risk factors, and PRSs achieve the best overall performance for myocardial infarction risk prediction, we modified the text to read:

We then modified both clinical feature space and model complexity variables together by training neural networks with established risk factors combined with large-scale diagnostic data, with and without also including PRSs (NG2; NN vs. NG2+PRS; NN). While the two neural network models utilizing large-scale diagnostic data had significantly higher recall (at class decision thresholds ranging from 0.5-0.99) compared to all other models (Fig. 2d and Fig. 2e), there was no significant difference in recall between the NG2+PRS; NN and NG2; NN models, with recall at class decision threshold 0.5 as high as 85% (+/- 1%) for both models. Even as the class decision threshold is increased to 0.8, the recall remains relatively high at 63-65% (+/- 3%).

To convey our primary point here without repeating the main results.

Q7) I do not understand the rationale for expecting a better added value of PRS to a model including more clinical information. Usually it becomes more difficult to improve the AUC of a model when good predictors are already in the model. Is there prior evidence to expect “potential effects of non-linear interactions between PRS and clinical features” (line 130)? I suggest to rephrase the title of the first section in the Results (lines 107-108).

These are excellent points.

We removed the sentence originally referenced in line 130.

This reduction in value-add was then compensated for when both large-scale clinical features and a neural network were used in the analysis (NG2; NN vs. NG2+PRS; NN), but the degree of compensation was minimal, nearly returning the value-add estimate to baseline ($\Delta\text{AUC}_{\text{PRS}} = 0.009 \pm 0.003$) rather than dramatically increasing the value-add. ~~as might be expected through the potential effects of non-linear feature interactions between PRSs and the large number of clinical features.~~

This is a typical assumption in machine learning that non-linear feature interactions should increase value-add, but we agree with your point, that it shouldn't be here in this particular case.

8) Were the PRS correlated with the features included in NG2? It would help to determine if some genetic information were captured from the medical history and learning algorithm.

The correlations among the NG2 factors and PRSs was near zero.

We added this sentence in the Further Analysis of Significant NG2 Diagnostic Features in Supplementary Results:

To build on our analysis of the feature correlations, we then looked at the correlations between features 235, 365, and 471 and the baseline non-genetic risk factor features (Supplementary Fig. 8). **The correlations with PRSs were all near zero, and all** were weakly correlated to many of the other features but feature 235 had the most correlations.

9) I suggest adding more information in the Discussion on the interpretability of the diagnostic features. Some of the elements in the Supplementary Discussion could be summarized in the main text. It seems for example that the top diagnostic features captured elements related to aging, sex, obesity smoking and lipids.

Thank you for your feedback. We have modified the Discussion section based on your feedback to incorporate these changes.

When we explored the ICD-10 diagnostic codes most important towards the meaning of diagnostic feature 235, we found that these codes were heavily enriched for conditions shown in the literature to be associated with increased myocardial infarction risk. For instance, hyperlipidemia (elevated blood lipid levels) is one of the most well-established risk factors for heart disease and plays a significant role in its development. Other prominent codes included those for obesity and type 2 diabetes, both of which frequently occur together and are strongly linked to increased heart disease risk. While not as well-established as risk factors, gastroesophageal reflux and gastritis—also relevant to the classification of node 235—have recently been identified as independent risk factors for heart disease.

Minor **points**

MP1) Please use the term myocardial infarction consistently (“heart attack” is a lay term).

This has been addressed throughout the manuscript. Apologies for this.

MP2) Please list the risk factors included in NG1 in the main text (Supplementary Table 7).

This information has been added to **Derivation of NG1 Feature Set (Small-Scale Disease-Specific Risk Factors)** in the **Methods** section.

Reviewer #3 (Remarks to the Author):

In this manuscript, the authors report the results of a study based on UK Biobank data aimed at developing a myocardial infarction risk prediction score. The novel aspect is the addition of polygenic risk scores to algorithms that include either a set of traditional cardiovascular risk factors or a broad array of information obtained from electronic health records. The authors conclude that addition of a polygenic risk score (PRS) to predictive models that are based on demographic and clinical data only minimally adds to risk prediction, independent of the modeling strategy.

Q1. The manuscript is well written and provides extensive data in support of the final conclusions. However, the use of the area under the curve (AUC) as the only model diagnostic criterion has its limitations. Although, there is admittedly no perfect way of deriving risk prediction models, the authors may want to consider examining model calibration and net reclassification, in addition to the AUC.

We thank you for taking the time to review our work and providing feedback.

As you mentioned, AUC is the most prevalent metric we use, as we are doing binary classification. AUC provides a clear and well-established measure of model performance, allowing us to compare the performance of our logistic regression and neural network models. Given the score of our study, AUC is a reliable metric for assessing overall performance in distinguishing between positive and negative cases.

However, while we did not use net reclassification, we did evaluate the models on several other metrics, such as recall, precision, and determination of incidence rates, as shown in **Figure 2**. We also show analyses expanding beyond the use of AUC in **Supplementary Table 1**.

We do agree that NRI is an emerging technique that is used as an alternative to AUC, and while we have used it in some other work, we were hesitant to do so in this paper. One reason is that AUC is still the de facto measure in most machine learning papers. Also, work by Pepe et al. has shown that NRI can be positive even when additional features add no predictive information when using large datasets, such as the one we have.

Q2. A flow chart showing inclusion/exclusion criteria would have been helpful. This chart should also indicate loss to follow-up at 10 years.

We completely agree.

We have added this as **Supplementary Figure 10**.

Q3. Use of the ICD10 codes in the development of the risk prediction models underutilizes the potential of EHRs. Provider notes can be messy and natural language processing protocols are not straightforward, however information contained in those notes can go significantly beyond diagnostic codes, which vary greatly in performance.

We agree with your position on NLP, and have published several papers using NLP in predictive tasks.

However, for this study, the needed data was not available to us in the UK Biobank dataset - we only had access to ICD-10 codes at the time of the study. While the performance of the EHR-based model would potentially benefit from inclusion of text-based NLP data, unfortunately, we could not perform this analysis.

Additionally, while having even better performance from the EHRs would be great, it does not change our conclusion that the EHR-based model achieves the best performance compared to the other models and that PRSs provide minimal value-add in comparison.

This is a great point, and hopefully something we can explore in the future.

Q4. It would have been helpful if the authors differentiated between EHR for different types of services (ambulatory care, hospitalizations, etc.). This would have added to the clinical utility of the EHR-based scores.

Thank you for your feedback.

Sorry for the confusion on this. We did include codes from EHR data in general practice records, hospital records, as well as death registry, cancer registry, and questionnaires. Please see the first sentence in the **Cohort Definitions** section in **Methods**:

Individuals were classified based on whether they had a given ICD-10 billing code in their medical record. Five diagnostic datasets available in the UK Biobank were used to identify cases: primary care records, hospital inpatient records, cancer registry data, death registry data, and self-reported questionnaire data (i.e., patients being asked about their medical history).

We also discuss this in the **Electronic Health Record (EHR) Representation Learning (Derivation of Non-Genetic Feature Set 2; NG2)** section; we did include codes from HER data in general practice records, hospital records, as well as death registry, cancer registry, and questionnaires.

Please let us know if you would prefer this information moved to another section, and we apologize for any confusion.

Q5. I was not convinced by the rationale to use multiple PRSs in one model. Although considered a priori, this further approach complicates interpretation of an already complex analysis. Separate analyses with the different PRSs would have been more transparent and justifiable.

This is another great question.

We used multiple PRSs as collectively they could capture slightly different independent components of genetic risk. While the different PRSs do correlate with each other between 0.12-0.68, as shown in **Figure 1c**, as also shown in in the same figure, the univariate AUCs for each PRS are all within the range of 0.53 – 0.56, with the combined PRS univariate measure only slightly higher. Thus, while there is some correlation among the PRSs, there is also sufficient evidence that each PRS may capture unique genetic information associated with myocardial infarction that can provide unique predictive information that enhances overall model performance.

Response to the Reviewers (Round 2)

General Response:

We again thank the reviewers and editorial board for their insights and attention to details. In response, we have included corrections and explanations for their further consideration. We are now pleased to re-submit the manuscript titled “Machine learning for myocardial infarction risk prediction: greater value-add from electronic health records than polygenic risk scores” with the corrections included.

As per the editorial comments, a revised manuscript should (1) address the remaining concerns of referees #2 and #3 in the manuscript, and (2) emphasize and discuss the clinical relevance/novelty (i.e. adding polygenic risk scores (PRSs) results in only a marginal improvement in predictive performance for myocardial infarction risk compared to models incorporating clinical and EHR-derived data alone).

The former is addressed in the comments and edits described below. The latter is addressed by adding one paragraph to the end of the paper, to better reflect both the clinical relevance and novelty of this study:

Altogether, our study represents, to our knowledge, one of the most comprehensive and thorough explorations of the value-add of PRSs over clinical data alone for myocardial infarction risk prediction, to date. On a practical level, the results of this study can provide value for practitioners and healthcare centers deciding how and in what contexts to integrate PRSs into clinical practice for risk assessments. Furthermore, it can provide a baseline for future studies and discussions that explore this question in different populations and clinical settings.

In this document, we include excerpts of the [revised text \(blue\)](#) presented in the context of original unchanged text (gray) for ease of reading and understanding the textual context in which the changes were made.

Reviewer #2 (Remarks to the Author):

The authors addressed most of my comments adequately. I have a few remaining remarks:

1) Some confusion remains regarding the number of individuals included and the exclusion criteria. The authors now indicate:
The 10-year incident case and control counts were 10,918 and 455,775. A total of 35,948 individuals who did not self-report as being of White British ancestry or who were lost to follow-up in the incident prediction time period, as per data field 191 and the death registry, were excluded from the analyses unless they had an incident diagnosis of the disease (Supplementary Table 5).

Does this mean that individuals of all ancestries were included if they had an incident myocardial infarction? Also, the total number of individuals when adding the prevalent cases and controls in Supplementary Table 5 seems to include all ancestries.

We apologize for the confusion and thank the reviewer for their thorough check of our response and attention to detail. To confirm, we excluded individuals of White British ancestry, and noticed that the text in Supplementary Table 5 required a correction.

The overall exclusion criteria are listed in the “Methods” section under “Dataset & Study Population” and summarized below:

- Participants with no medical history prior to their first assessment visit
- Participants with no available genotype data
- Participants who opted to be removed from the UK Biobank were excluded from the analysis
- Participants who were not self-reported White British ancestry

Specifically for incident cases for different follow-up time windows, the criteria detailed under “Cohort Definitions” are additionally relevant and constitute the “Excluded” cases in Supplementary Table 5.

We have updated Supplementary Table 5 and Supplementary Figure 11 with the corrected numbers.

Additionally, we have updated the text under “Cohort Definitions” in the “Methods” section as follows:

To establish cohorts for incident disease risk prediction, we used the year of each participant’s first UK Biobank assessment visit as the baseline time point for that participant and excluded all participants with a pre-existing diagnosis prior to their assessment visit. **The 10-year incident case and control counts were 7,515 and 315,939, respectively. A total of 179,187 individuals were excluded from the analyses based on the exclusion criteria outlined in Supplementary Table 5.**

2) The selection of the outcome as myocardial infarction yes/no over a follow-up period of 10 years while excluding all events occurring before recruitment could favor predictors of late-onset events. PRS have been shown to better predict early-onset events, which might explain their lower performance in the current study. I suggest to discuss this point and/or add as a limitation.

In our study, we did exclude all participants with a prior myocardial infarction before recruitment into the UK Biobank, as our focus was to predict first-time incident myocardial infarction. As such, any participants with a high genetic risk that had already experienced their first myocardial infarction, if applicable, would be excluded from the analysis, potentially skewing the study population to include participants with lower genetic risk, and thus reducing the overall performance of the PRSs. This is likely reflected in the slightly lower PRS AUCs of when looking at prevalent cases (AUCs approximately 0.6) vs. 10-year incident cases (AUC = 0.582 +/- 0.01).

The best way that we could have addressed this would have been to look at 10-year incident myocardial infarction risk, independently of whether this is first-time, second-time, etc. This would have included any participants that had their first event prior to their first day of recruitment *and* had a subsequent second event during the 10-year window as cases; while those with their first event prior to recruitment and no subsequent second event would be included as controls. It would essentially involve moving some of the ~35k excluded participants (<8% of total) back into the study, if they had a diagnosis during the 10-year window. Some of those participants might have had higher PRSs for myocardial infarction, which would then potentially increase the value-add of the PRSs as per the study results.

While our study explicitly focused on first-time 10-year incident myocardial infarction prediction, it is a valid point that focusing on the broader question irrespective of first-time incident myocardial infarction might lead to a

different result, i.e., stronger performance of the PRSs. Arguably, this would also lead to much stronger performance of the EHRs, as having had a previous myocardial infarction event is one of the strongest predictors of a future event, independent of genetic risk (i.e., Jernberg et al. 2015, *European Heart Journal*).

To emphasize this more, we have modified the 3rd to last paragraph of the Discussion:

Our study also focused on predicting first-time 10-year incident myocardial infarction and excluded individuals that had already experienced a documented heart attack prior to the start of the incidence window. Including these individuals, i.e., by evaluating risk of 10-year incident myocardial infarction risk independent of prior diagnosis with the condition, might have modified the results of our study in several ways. First, having a high genetic risk for myocardial infarction increases the chances that an individual will have their first event early, and as such, including these individuals from the study might skew the study population to include participants with relatively higher genetic risk, thus increasing the overall performance of the PRSs. Second, including participants that already had their first event prior to the start of the study would likely result in much stronger performance of the EHR-based feature set in our study, given that having had a previous myocardial infarction event is one of the strongest predictors of a future event, independent of genetic risk. Thus, if we did explore that question, this likely would not have affected our key results, or might even further dilute the predicted value-add of PRSs. Nonetheless, future work is warranted to evaluate this empirically. ~~Including these individuals might have resulted in improved prediction for the clinical data only models, given that a history of a heart attack is generally a good predictor of a later one. However, this likely would not have affected our key results, or if anything it might further dilute the predicted value-add of PRSs. In addition to this, future work is warranted to adapt this approach towards predicting incident coronary artery disease in individuals without a prior history of any heart-related conditions, or for other conditions.~~

3) Regarding the transferability of this method to different populations and settings, I still believe that the information/performance of the EHR could differ according to population-dependent factors such as healthcare organization, prevalence of disease and risk factors, even cultural differences among the same ethnic group (European ancestry in this case).

We thank the reviewer for pointing this out again and here provide a more thorough response than in our previous comments.

The transferability issue is a critical consideration when evaluating the broader applicability of our work in this study, especially given that we are using diagnostic codes that are subject to variability across different populations and settings. Our previous work (Isgut et al. 2023, *Genetic Epidemiology*) studied the role of data sources from which cases and controls are selected on subsequent results from genome-wide association studies (GWAS) and found significant variability between data sources for some diseases. For example, for iron deficiency anemia, we found that cases defined using just hospital inpatient ICD-9 and ICD-10 codes from the UK Biobank were significantly different genetically (as shown by genetic correlation, r_g) when compared to cases defined using codes from a primary care setting. While this was not relevant for all diseases we evaluated (i.e., relative homogeneity of results for glaucoma), this further highlights the variability that can occur when considering diagnostic data provided from different sources.

The EHR-based features extracted for the NG2 dataset were clinical diagnostic codes across UK hospital inpatient, primary care, cancer registry, and death registry datasets, as well as from the UK Biobank self-reported clinical diagnostic codes dataset, as discussed in Supplementary Fig. 3 and in the Supplementary Text. As such,

the data used in our work is not limited to any specific healthcare organization, city, region in the UK, or category of data source (i.e., hospital inpatient data, or just primary care, etc.). Nonetheless, it is true that the data used will be specific to a) the European ancestry population in the UK Biobank, and b) more broadly, the European ancestry population in the UK. Given that the model used to derive the NG2 dataset extrapolates patterns in the diagnostic data, these will depend on the general frequencies of diagnoses and co-morbidities between diagnoses that occur in the UK Biobank. The UK Biobank population is mostly self-selected and may not entirely reflect the population of the UK, and the UK itself has cultural, ancestry, and other factors that differentiate it from other parts of the world.

Thus, we agree that this is a limitation of our study, and of other studies that only leverage diagnostic data from the UK Biobank, without consideration to population subgroups. In future work, perhaps simulations can be done by taking subsets of the UK Biobank population by ancestry, cultural or religious group, region, etc., and comparing the results of this study between groups. Alternatively, other future work might evaluate other population datasets, such as the Million Veteran Program (MVP), FinnGen, etc. However, that is out of scope of the current work.

The transferability of the results from this study to other populations can depend on broad factors, such as the frameworks and established practices by which diagnoses are made by clinicians in different settings and in different specialties, differences in patterns of comorbidities that occur across population groups, cultural factors that impact patient-clinician interactions, and much more. As such, it is difficult to say exactly how broadly translatable these results are, but at best, our results provide a foundation for future work that can adapt similar frameworks to explore the value-add of PRSs over EHR-related data in other populations.

To reflect this limitation, we will change the 5th to last paragraph of the Discussion, as such:

There are several limitations of our approach. First, we focused our analysis on individuals of European ancestry only given that this was the most highly represented ancestry group in the UK Biobank dataset. ~~Future studies should evaluate this question in the context of different ancestry groups to confirm whether the results are generalizable.~~ Furthermore, the UK Biobank is mostly a self-selected population and is limited to individuals in the UK, which has cultural, ancestry-related, and other factors that differentiate it from other parts of the world. As EHR-based representation learning algorithms extrapolate patterns in the diagnostic data, these depend on the general frequencies of diagnoses and co-morbidities between the diagnoses that occur within the UK Biobank, which may not translate to broader populations. The same can be said about our 10-year incident myocardial infarction risk prediction models. Future work should adapt these frameworks to other ancestry groups and populations using datasets such as the Million Veteran Program (MVP), All of Us, FinnGen, and others, to explore these findings in broader contexts.

4) Regarding the correlation between PRS and EHR features, the statement “The correlations with PRSs were all near zero” is imprecise, I suggest to add the PRS to the variables reported in Supplementary Figure 8 to support it.

Thank you for noting this.

Additionally, we have created a separate heatmap (now **Supplementary Fig. 10**) with just the three selected NG2 features (235, 365, and 471) from **Supplementary Fig. 8**, comparing them to the PRS features. We use

data from just the 10-year incident cases and controls, as in the original work. As shown, the correlations range from -0.01 to 0.04.

To clarify the description in the Supplementary Results under “Interpretation of Diagnostic Features – Additional Results” and “Further Analysis of Significant NG2 Diagnostic Features”, we will revert the last paragraph to its original form and add a few sentences at the end to reflect the PRS correlation analyses:

To build on our analysis of the feature correlations, we then looked at the correlations between features 235, 365, and 471 and the baseline non-genetic risk factor features (**Supplementary Fig. 8**). All were weakly correlated to many of the other features but feature 235 had the most correlations. It was most correlated with 365 (Pearson $r^2 = 0.26$), followed by risk factor features body mass index (BMI) (Pearson $r^2 = 0.18$) and cholesterol (Pearson $r^2 = 0.16$), which is supported by our findings on some of the top codes associated with diagnostic feature 235 (i.e., related to hyperlipidemia and obesity). We do not provide extensive analysis details on the other features because they each independently add minimal contribution to prediction and are difficult to interpret. We also looked at the four PRSs to explore correlations between these and the three NG2 features evaluated (235, 365, and 471). As shown in Supplementary Fig. 10, these range from -0.01 to 0.04 and are thus negligible, suggesting that genetic information is not significantly captured from the medical history machine learning algorithm.

Reviewer #3 (Remarks to the Author):

I appreciate the exhaustive responses provided by the authors to the initial critique. However, several remaining aspects need to be addressed.

1. The overall premise of this article is that PRS provide added value to myocardial infarction (MI) risk prediction models. However, study results suggests that this value is minimal. Additionally, the stated utility of PRS in clinical settings is impractical, at best. Therefore, contrary to the authors’ concluding statements, study findings do not support an increasing role of PRS in clinical practice.

Our concluding statements in the last paragraph of the Discussion are as follows:

“Overall, through our systematic analysis of the value-add of PRSs in the context of varied model complexity and clinical feature sets, our results suggest that using large-scale EHR data in neural networks is a promising avenue towards improving myocardial infarction risk prediction and provides greater value-add for prediction than using polygenic risk scores. As EHR deep representation learning models become more sophisticated, they might potentially be better able to account for gene-environment interactions, and there may be an increasing role for PRSs in clinical practice. However, more research is needed to further elucidate the clinical utility of PRSs and their potential value-add in different clinical contexts.”

We believe the reviewer might be referring to the following sentence: “As EHR deep representation learning models become more sophisticated, they might potentially be better able to account for gene-environment interactions, and there may be an increasing role for PRSs in clinical practice.”

To clarify, this statement is intended to provide a positive, optimistic outlook to readers on the potential future value-add for PRSs, if for example, gene-environment interactions were to be better leveraged in these models. As gene-environment interactions are well documented in the literature, and can be taken into account in deep learning models that include both genetic and EHR-derived features, there might be the chance that these models achieve better performance. In other words if gene-environment interactions are better captured in future prediction models, then adding genetic data might improve on prediction results from EHRs alone. This is more likely to be better captured by deep learning models that include individual genotypes (i.e., SNPs) as inputs in addition to diagnostic codes, and that include more comprehensive attention-based, residual, or other components that might better leverage and interpret patterns representing these interactions. This would likely require significantly larger population-level datasets than what is currently available but could one day result in a better value-add from the inclusion of genetic data.

As such, we still abide by this statement in the conclusion and thank the reviewer for the opportunity to clarify what we meant by this. To clarify the language, we have modified this paragraph as follows:

Overall, through our systematic analysis of the value-add of PRSs in the context of varied model complexity and clinical feature sets, our results suggest that use large-scale EHR data in neural networks is a promising avenue towards improving heart attack risk prediction and provides greater value-add for prediction than use of polygenic risk scores. Although genomic data minimally improved on 10-year risk prediction performance over EHR data in our study, it is worth noting that there may still be untapped value in the multi-modality integration of large-scale EHR-based and genomic data in advanced deep learning models. Gene-by-environment interactions are well documented in the literature, so it follows that there may be the potential for greater improvements in performance from the multi-modality approach. For example, as deep learning models become more sophisticated in the future, and as larger population-level datasets become available that include both genomic and EHR data, it is plausible that interaction effects, if they exist, might be better captured, lending additional value-add to the notion of including PRSs in 10-year myocardial infarction prediction models. Data inputs that might better enable this include leveraging different genomic features (i.e., genotypes rather than PRSs as univariate scores), and disease prediction models might be more advanced than those used in this study, leveraging attention or residual components, for example, that might better capture patterns representing these interactions. Future work is warranted to build on this analysis to explore the question in broader contexts. ~~As EHR deep representation deep learning models become more sophisticated in the future, with more available data to enhance large-scale model training and performance, they might potentially be better able to account for gene-environment interactions, and thus demonstrate a greater value-add from integrating genomic data in risk prediction models. there may be an increasing role for PRSs in clinical practice. However, more research is needed to further elucidate the clinical utility of PRSs and their potential value-add in different clinical contexts.~~

2. Please provide information on how do the results based on the NG1 model only compare to existing MI risk prediction models.

The first-time 10-year incident myocardial infarction (MI) risk prediction model with NG1 alone achieves an AUC of 0.717 $\pm$ 0.007 (**Supplementary Table 8**), and leverages the features described in **Supplementary Table 7**.

It is difficult to compare this with the large number of other clinical data only MI risk prediction models that have been published to date, given the different performance metrics used by different studies, the different

populations used (i.e., varying by country, ancestry, sex, pre-existing conditions, and other demographic factors), the different time windows for prediction (i.e., 30-day event prediction as per O’Rielly et al. 2023, 10-year prediction as per various studies including the Framingham Risk Score (FRS), ASCVD risk score, etc.), and the clinical definition of MI cases used in the study. This complication is further evidenced by the meta analysis by Damen et al. 2016, *BMJ*, which looked at 212 studies describing clinical models to predict cardiovascular disease (myocardial infarction being a subset) and notes substantial heterogeneity in predictor and outcome definitions, and more. As a comparison point, the FRS is one of the more established 10-year scores (although it looks at heart disease overall), and achieves an AUC of 0.73 (CI from 0.71 to 0.75) in a 2005 prospective study of ~7k males aged 40-59 (see “Metabolic Syndrome vs. Framingham Risk Score for Prediction of Coronary Heart Disease, Stroke, and Type 2 Diabetes”, *JAMA Internal Medicine*). Again, as this population is different from that of the UK Biobank in several ways (age, sex, etc.), it is not the best comparison. Another study by Alaa et al. 2019, *Plos One*, found that the FRS achieved an AUC of 0.724 +/- 0.004 in the UK Biobank, but focused on 5-year prediction, and looked at incidence of more general cardiovascular events rather than just myocardial infarction (i.e., included cases with newly diagnosed heart failure, cerebrovascular disease, vascular dementia, ischemic heart disease more broadly, etc.). They also only used hospital inpatient data.

To achieve a reliable comparison point, we would have to calculate the FRS ourselves on the same case and control participants in the UK Biobank that we used in the original study. As this is outside the scope of our study, and would be a significant undertaking to ensure robustness, we did not do this.

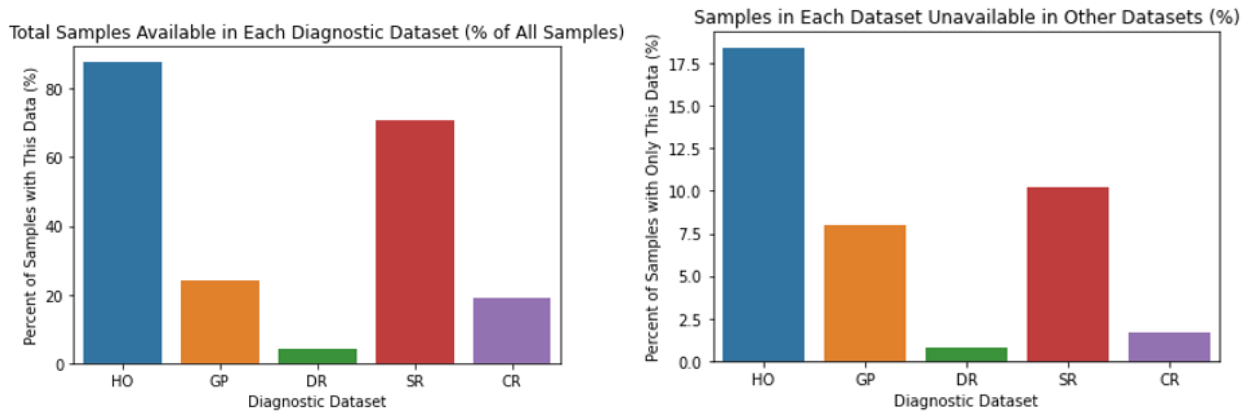
For the NG1 feature set, we looked at a reasonable combination of lab test results (cholesterol, triglycerides, and lipoprotein A), physical measurements (systolic blood pressure, BMI), baseline demographic features (age at recruitment, sex), lifestyle factors (smoking status), and family history of heart disease, which have been accepted as inputs for prediction models in our previous published work on heart disease risk prediction (i.e., Isgut et al. 2021, *Genome Medicine*; Isgut et al. 2024, *Scientific Reports*)

Thus, we are confident that our NG1 model performance reflects what would reasonably be expected as per the prior literature.

3. The authors may have mis-interpreted the initial request to consider as independent predictors in the expanded NG2 model, ICD codes for encounters that occurred in different clinical settings. In other words, there is clinical relevance to risk prediction from just ambulatory care data, in the absence of evidence of hospitalizations (within a defined time period). The same would be true for ICD codes obtained solely from Emergency Department visits.

We apologize for the previous interpretation of your comment. This feedback relates to the distinction between different sources of the clinical data we obtained from the UK Biobank to train the model used to derive our NG2 features. We leveraged data from a diversity of data sources available in the UK Biobank dataset, as shown in **Supplementary Fig. 3**, including hospital inpatient, primary care (i.e., ambulatory care), cancer registry, death registry, and self-reported questionnaire data provided at recruitment to the study. Our rationale for including all of these data sources was to maximize the available diagnostic data for each patient in the UK Biobank. While some participants, for example, might have diagnoses obtained from the hospital, others might have prior diagnoses obtained from primary care settings only.

Furthermore, some participants might not even have any diagnoses reflected in their record, were we to consider just one data source, as shown in the figures below from one of our previous analyses:



HO = hospital inpatient, GP = general practice or primary care, DR = death registry, SR = self-reported, and CR = cancer registry. **Note: this figure has not been added to the paper, but is included here for explanation purposes.**

As such, our objective for this study was, in general, to maximize the amount of diagnostic data available for training.

Nonetheless, we note that data sources have a significant impact on the resulting ICD-10 codes. In our previous work (Isgut et al. 2023, *Genetic Epidemiology*), we showed that cases for the same condition defined from different care settings resulted in differential results in subsequent genome-wide association studies (GWASs). For example, iron deficiency anemia cases, as defined by looking only at hospital inpatient records in the UK Biobank, had a significantly different genetic correlation (r_g) than cases defined from general practice data only.

Further exploration of the role of specific data sources might involve a study design where we build the representation learning models using individual data sources only. For example, we might in that case train the same model at different times to learn NG2 features using just hospital inpatient data, just general practice data, etc., and then compare the value-add of PRSs with each of the different representative NG2 datasets. This would require extensive work that is out of scope of the current study, but we believe it could adequately address the question of, “what is the role of different types of clinical data sources in the broader implications of these results?”

To address this comment, we have added the following to the second-to-last paragraph of the Discussion section:

Finally, as this study broadly uses diagnostic data from various sources available from the UK Biobank, future work might further explore this question in the context of specific clinical settings, such as in comparison between ambulatory and in-patient care settings, to explore to what degree the results of the study vary in these different contexts.